the viruses, which differ profoundly their structural and biological properties. Microbiology is without aquestion, a branch of biology that possesses both unity and coherence.

The following list of specializations in microbiology will provide an insight on the significance of microbiology in the world today:

- Medicine
- Environmental science
- Food production
- Fundamental Research
- Agriculture
- Pharmaceutical Industry
- Genetic Engineering

The popular perception among the general public, however, remains one of infections and plagues. In reality, only a couple of hundred out of the half million or so known bacterial species, termed as pathogens with the potential to cause disease give rise to infections in humans and dominate the microbial world.

1.1 Microbes in Space

The majority of experiments on microorganisms in space were performed using Earth-orbiting robotic spacecraft, Example: the Russian ***Foton*** satellites and the ***European Retrievable Carrier*** (EURECA) (121), or human-tended spacecraft, such as space shuttles (106, 107) and space stations, Example: ***MIR*** and the ***International Space Station*** (ISS).

Only twice, during translunar trips of ***Apollo 16*** and ***17*** in the early 1970s, were microorganisms exposed to space conditions beyond Earth's magnetic shield, in the MEED (microbial ecology equipment device) facility and in the Biostack experiments. Arriving in space without any protection, *microorganisms* are confronted with an extremely hostile environment, characterized by an intense radiation field of galactic and solar origin, high vacuum, extreme temperatures, and microgravity.

HOTS

1. What are the sources of microorganisms in space?
2. How do bacteria survey in space?

Some bacteria were found in International Space Station and on the Mars Rover. Some bacteria and tiny microbes called tardigrades are able to survive for longer periods in space. This ability of surviving in extreme environmental condition leads to forward contamination. Sea planktons and other microorganisms have been identified in cosmonauts' spacewalk samples.

In July 2016, Kate Rubins was the first to sequence DNA in space. NASA astronaut Peggy Whitson amplified and sequenced the DNA of bacteria that grew as colonies in the petri plate on the surface on the space station. In June 2018, Professor George Fox and his team have isolated genus *Bacillus* from spacecraft assembly rooms at the Jet Propulsion Laboratory. They have sequenced the complete genomes of two strains, B. *safensis* FO-36bT and B. *pumilus* SAFR-032 and found that they are resistant to radiation.

Los Angeles in great news for India, scientists at NASA have named a new organism discovered by them after the

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